Aman Mehtar

AI Engineer & Full-Stack Developer

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Summary

AI & Data Science engineering student who builds agents, developer tools, and full-stack products in public. Trains language models from scratch on consumer hardware, ships real-time applications, and self-hosts infrastructure end to end — from pretraining and quantization to React frontends and Go services.

Education

University of Mumbai — Mumbai, India

Bachelor of Engineering in Artificial Intelligence & Data Science — *2023 – 2027*

Experience

AI / LLM Engineer Intern — Jun 2024

contexIQ — *Remote*

- Built an LLM-powered e-commerce semantic search system — Llama on Groq for intent parsing, Weaviate for cosine vector retrieval — a winning project of the internship.

Development Coordinator & AIML Associate — Aug 2024 – Present

Microsoft Learn Student Club (MLSC), VCET — *Mumbai, India*

- Led a team of 4 in the Amazon ML Challenge 2024, coordinating end-to-end model development.
- Designed and deployed an FAQ chatbot for the MLSC Coherence Hackathon site.
- Coordinated MLSC hackathon operations and contributed to multiple industry initiatives.

Robotics Intern — Jun 2021 – Sep 2022

ScienceKidz Educare Ltd. — *Mumbai, India*

- Built obstacle-avoiding robots using IoT sensors and embedded systems (Robo-Doctor, Obstacle Avoiding Car).
- Developed an Android logistics-manager app to optimize company orders and delivery.
- Designed components with CorelDRAW and ran MDF & acrylic laser-cutting operations.

Projects

LLM & chessLLM | *Python, PyTorch* — GitHub

- Pretrained a 101M-parameter LLM from scratch on an 8 GB laptop GPU.
- Built chessLLM, a 94M model that forms an internal board from move text alone — linearly probeable to 85.6% accuracy.

jcow411 — Agent-Orchestrated Firmware Repurposing | *Claude Code, DeepSeek, Gemini, Antigravity* Writeup

- Orchestrated two autonomous coding agents (DeepSeek V4.1 via Claude Code, Gemini 3.8 Flash via Antigravity) to root and repurpose a decommissioned gateway into a self-healing Wi-Fi repeater — executed end-to-end by the agents.
- Adjudicated a confident but wrong "impossible" verdict: the first agent misread a vendor error code, so cross-checking against a second model surfaced the true root cause and produced a working solution — a multi-agent, human-in-the-loop workflow.

amzn-ml-2025-pricing | *Qwen, 4-bit LoRA, PyTorch* — GitHub

- Amazon ML Challenge 2025 price-prediction entry using 4-bit LoRA on Qwen with a monotone quantile head, trained on a single laptop GPU.

Phi-3.5-Vision Finetuning | *Phi-3.5-Vision, PyTorch* — Kaggle

- Amazon ML Challenge 2024: finetuned a VLM to extract product attributes (weight, quantity) from sticker images, with normalization rules reaching 95% accuracy.

quick-gateway | *Docker, Python, LiteLLM, Vertex AI* — GitHub

- Docker proxy bridging Claude Code, VS Code, and OpenCode to Vertex AI models (GLM, Qwen, DeepSeek, Kimi) with token refresh, cost tracking, and request logging.

ApexCore | *React Native, TypeScript, llama.cpp* — GitHub

- Offline on-device Android AI running quantized SLMs (Llama, Phi, Qwen, Gemma) via llama.cpp, with Hugging Face model browsing, benchmarking (tokens/sec, latency, memory), and custom personas.

Sentira | *Next.js, FastAPI, Gemini, PostgreSQL* — GitHub

- Full-stack platform using Gemini to analyze video, audio, and text via parallel LLM pipelines, with real-time sentiment tracking, bias detection, claim verification, and exportable PDF reports.

Image Generation Studio | *Next.js, Vertex AI, WorkOS, AWS S3* GitHub

- Prompt-based image generation on Vertex AI with WorkOS auth, S3 storage, drag-and-drop nested folders, multi-provider support, and cached presigned URLs.

TrackMate — Pet Tracking Collar | *ESP32, LoRa, NEO-6M GPS* GitHub

- Long-range, low-power pet tracker with firmware collecting GPS over LoRa to desktop/mobile clients as a reliable end-to-end system.

Technical Skills

ML / LLMs: From-scratch pretraining & post-training, LoRA fine-tuning, quantization, evals, llama.cpp
Languages: Python, TypeScript, Go, C/C++, Bash
Full-Stack: React 19, Next.js, React Native (Expo), FastAPI, Convex, OAuth, PostgreSQL / Drizzle
Systems: WebSockets / SSE, low-latency APIs, Docker, self-hosting, native Linux tooling
Vision & AI: YOLO, OpenCV, multimodal LLMs (Gemini, Phi-3.5-Vision), semantic / vector search

Volunteering

Technical Volunteer Aug 2024 – Present
National Service Scheme (NSS), VCET *Mumbai, India*

- Designed social-media creatives to grow NSS VCET's presence; joined community drives (beach clean-ups, Independence Day).